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Moisture assessment system for wound care

Abstract

A Wound Moisture and Assessment System includes a flexible coupon containing one or more electrochemical moisture sensors for measuring a moisture profile of a wound. The coupon is activated by a wireless RF scanner that provides power to the coupon for taking moisture measurements, receives the moisture measurements therefrom, and transmits the measurements to a computing device. The moisture measurements can be represented as a wound moisture map that allows a patient or caregiver to assess the health and healing progress of a wound such as an ulcer.

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References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6433244	12/2001	Roe et al.	N/A	N/A
6720866	12/2003	Sorrells et al.	N/A	N/A
6774800	12/2003	Friedman et al.	N/A	N/A
7456744	12/2007	Kuhns et al.	N/A	N/A
7948380	12/2010	Kuhns et al.	N/A	N/A
8978452	12/2014	Johnson et al.	N/A	N/A
9070060	12/2014	Forster	N/A	N/A
9506886	12/2015	Woodbury et al.	N/A	N/A
9526439	12/2015	Connelly et al.	N/A	N/A
9649230	12/2016	Li	N/A	N/A
10022277	12/2017	Heil	N/A	A61F 13/42
2004/0133143	12/2003	Burton	602/58	A61F 13/42
2007/0021074	12/2006	Posamentier	N/A	N/A
2008/0061965	12/2007	Kuhns et al.	N/A	N/A
2013/0178812	12/2012	Flach	604/385.03	A61F 13/0203
2013/0271278	12/2012	Duesterhoft et al.	N/A	N/A
2013/0274563	12/2012	Duesterhoft et al.	N/A	N/A
2013/0274629	12/2012	Duesterhoft et al.	N/A	N/A
2014/0005618	12/2013	Locke et al.	N/A	N/A
2014/0027295	12/2013	Chiao et al.	N/A	N/A
2014/0066868	12/2013	Freedman et al.	N/A	N/A
2014/0200538	12/2013	Euliano et al.	N/A	N/A
2014/0298928	12/2013	Duesterhoft	73/865.8	A61M 1/73
2015/0018792	12/2014	Marsiquet	604/361	A61F 13/42

2016/0166438	12/2015	Rovaniemi	N/A	N/A
2016/0267769	12/2015	Rokhsaz et al.	N/A	N/A
2017/0048651	12/2016	Lin Charna	N/A	A61F 13/42
2017/0065464	12/2016	Heil	N/A	G06K 7/10366
2018/0055359	12/2017	Shamim	N/A	A61B 5/0004
2021/0148836	12/2020	Hameed	N/A	H01Q 1/225

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101049261	12/2006	CN	N/A
3034054	12/2015	EP	N/A
20170113930	12/2016	KR	N/A
2010011920	12/2009	WO	N/A
2012054474	12/2011	WO	N/A
2016166731	12/2015	WO	N/A

OTHER PUBLICATIONS

Mehmood, N., et al., Applications of modern sensors and wireless technology in effective wound management, J Biomed Mater Res Part B, 2013, 1-11. cited by applicant

Mehmood, N., et al., A flexible and low power telemetric sensing and monitoring system for chronic wound diagnostics, Biomed Eng Online, 2015, 1-12. cited by applicant

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Background/Summary

TECHNICAL FIELD

(1) This disclosure relates to systems and methods for monitoring wound moisture and exudate levels. The disclosed systems and methods can be used to guide the treatment of wounds by clinicians in a hospital or similar setting or by patients opting to self-manage their wounds.

BACKGROUND

(2) Chronic wounds affect approximately 4 million people annually in the United States. With a growing elderly population, increasing incidence of diabetes, venous stasis ulcers, and pressure ulcers (PUs), along with rapidly rising wound care costs, wound care is a significant public health problem. It is estimated that at least 1.7 million people develop pressure ulcers annually, at a cost of \$2.2 to \$3.6 billion (Langemo, D K, The Lived Experience of Having a Pressure Ulcer: A Qualitative Analysis Adv Skin WoundCare; 13:225-35; 2000). Venous stasis ulcers also present a major health management challenge for health care professionals. It is estimated that approximately 1% of the general population and 3.5% of persons over age 65 have venous stasis ulcers with numbers rising as the population ages (Hess, C. T., Management of the Patient with a Venous Ulcer, Adv Skin WoundCare; 13 (2); March-April 2000; pp. 79-83). The recurrence rate of venous stasis ulcers is approaching 70% (Bryant RA, editor. Acute and Chronic Wounds. St Louis: Mosby; 1992 p. 164-204). The estimated per-episode cost of care can exceed \$40,000 (Rudolph, D. M. Pathophysiology and Management of Venous Ulcers. J Wound Ostomy Continence Nurs 1998;

25:248-55) with a total cost of treatment estimated to be \$2.5 to \$3.5 billion (Onegnae, K. Phillips, T. Leg Ulcer Management. Emerg Med 1993; 25:45-53).

(3) The ability of a body to self-heal a wound is dependent, in part on its ability to maintain optimum moisture levels throughout and immediately surrounding the wound bed. Too little moisture can lead to dryness, cracking and irritation; too much moisture, e.g., chronic exudate overproduction, can lead to infection and tissue breakdown. In many cases, wounds are covered with protective dressings to generally protect the wound bed itself; however, this sometimes precludes the ability to monitor the state of the wound including, in particular, the moisture level. Removing a dressing to examine a wound can itself lead to irritation and disruption of the healing process, especially if the dressing is intended to protect the wound for extended periods of time.

SUMMARY

(4) In one exemplary aspect, a coupon for wireless communication with a transceiver is provided. The coupon includes a flexible substrate having a top surface and a bottom surface, a radio-frequency antenna circuit disposed on the top surface of the flexible substrate, and at least one electrochemical moisture sensor circuit disposed on the bottom surface of the flexible substrate.

(5) In one embodiment, the coupon further includes a near-field communications module disposed on the top surface of the flexible substrate that is in electronic communication with the radio-frequency antenna circuit. In a related embodiment, the near-field communications module includes a microcontroller in signal communication with the radio-frequency antenna circuit and the at least one electrochemical moisture sensor circuit, and a memory for storing executable logic functions for measuring a moisture value of an environment proximal to the at least one electrochemical moisture sensor circuit. In a further related embodiment, the near-field communications module further includes a multiplexer in signal communication with the microcontroller and the at least one electrochemical moisture sensor circuit, a signal converter capable of converting analog signals to digital signals and vice-versa, and an optional temperature sensor.

(6) In one embodiment, the coupon further includes a bacteria sensor disposed on the bottom surface of the flexible substrate that is in signal communication with the near-field communications module.

(7) In one embodiment, the near-field communications module is configured to receive operational power from an external radio-frequency source that is received by the radio-frequency antenna circuit.

(8) In one exemplary aspect, a system for wirelessly obtaining a moisture measurement is disclosed. The system includes a wireless reader including a transceiver configured to emit and receive radio frequency signals and a coupon. The coupon includes a flexible substrate having a top surface and a bottom surface, at least one electrochemical moisture sensor circuit disposed on the bottom surface of the flexible substrate that is configured to determine a moisture value, an antenna circuit disposed on the top surface of the flexible substrate configured to at least receive the radio frequency signals from the transceiver, and a microcontroller in electronic communication with the antenna circuit and the at least one electrochemical moisture sensor circuit.

(9) In one embodiment, the system further includes a bacteria sensor circuit disposed on the bottom surface of the flexible substrate and configured to be in electronic communication with the microcontroller.

(10) In one embodiment, the microcontroller is configured to obtain the moisture value from the at least one electrochemical moisture sensor circuit. In a related embodiment, the microcontroller is configured to wirelessly transmit the moisture measurement to the wireless reader via the antenna circuit.

(11) In one embodiment, the moisture value is determined by obtaining a capacitance value of the at least one electrochemical moisture sensor circuit.

(12) In one embodiment, the coupon is disposed on, or within a wound dressing or bandage. In one

embodiment, the microcontroller is configured to be powered by the radio frequency signal emitted by the transceiver of the wireless reader.

(13) In one embodiment, the coupon includes an array of electrochemical moisture sensor circuits, each in signal communication with the microcontroller. In a related embodiment, the system further includes a multiplexer in electronic communication with the microcontroller that is configured for obtaining a capacitance value of each of the electrochemical moisture sensor circuits.

(14) In one exemplary aspect, a system for generating a wound moisture map is disclosed. The system includes a flexible coupon having a top surface and a bottom surface, wherein the bottom surface is configured to be applied at least in part to a wound or ulcer. The system further includes an antenna circuit and a microcontroller disposed on the top surface of the coupon, wherein the microcontroller is in electronic communication with the antenna circuit, and a plurality of electrochemical moisture sensor circuits disposed on the bottom surface of the coupon. Each electrochemical moisture sensor circuit of the plurality of electrochemical moisture sensor circuits is in electronic communication with the microcontroller, and the antenna circuit is adapted to send and receive radio frequency signals to and from a remote radio frequency transceiver, respectively.

(15) In one embodiment, the system further includes an electrochemical bacteria sensor disposed on the bottom surface of the coupon that is in electronic communication with the microcontroller.

(16) In one embodiment, the system further includes an electronic temperature sensor disposed on the bottom surface that is in electronic communication with the microcontroller.

(17) In one embodiment, the microcontroller is configured to receive query signals from the transceiver via the antenna, query the plurality of electrochemical moisture sensor circuits in a selective- or group-wide modality for one or more wound moisture measurements, and transmit the moisture measurements to the transceiver via the antenna.

(18) In one embodiment, each electrochemical moisture sensor circuit of the plurality of electrochemical moisture sensor circuits includes an array of interdigitated electrodes.

(19) Certain advantages of the systems and methods include: low cost fabrication polymer thick-film (screen-printing), disposability, body-worn comfort, lightweight construction (e.g., printing on a polymeric substrate), real-time measurement capability to aid clinical management of wounds, wireless communication, adaptability to several dressing types (e.g., transparent film, foam, composite), a noninvasive approach with the ability to assess moisture without dressing removal (thereby reducing risk to pathogens, decrease number of dressing changes and in turn strengthening new epidermal tissue), ability to map the pattern of moisture across the wound surface (e.g., measuring wound moisture distribution and time variation).

(20) Further advantages include the ability to quantify moisture imbalance in wounds, the ability to administer timely and effective management of exudate levels, the ability to integrate wound moisture assessments into Electronic Medical Records systems for a variety of reasons, including specialty wound clinics for outcomes research, the ability to provide a reimbursement strategy for use of moisture measurement based on objective wireless evidence and integration of wound management in telecare or online decision support systems, especially for remote rural settings and managed care homes.

(21) Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art. Although methods and materials similar or equivalent to those described herein can be used in the practice or testing of any described embodiment, suitable methods and materials are described below. In addition, the materials, methods, and examples are illustrative only and not intended to be limiting. In case of conflict with terms used in the art, the present specification, including definitions, will control.

(22) The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features will become apparent by reference to the drawings and the following detailed description and claims.

Description

DESCRIPTION OF DRAWINGS

- (1) The present embodiments are illustrated by way of the figures of the accompanying drawings, which may not necessarily be to scale, in which like references indicate similar elements, and in which:
- (2) FIG. 1 illustrates a wound moisture assessment system (WMAS) according to one embodiment;
- (3) FIG. 2 is a functional diagram of a WMAS according to one embodiment;
- (4) FIG. 3 illustrates a WMAS coupon according to one embodiment;
- (5) FIG. 4 illustrates a moisture sensor circuit according to one embodiment;
- (6) FIG. 5 illustrates a circular loop air coil antenna according to one embodiment;
- (7) FIG. 6 illustrates frequency and bandwidth parameters of a moisture sensor circuit;
- (8) FIG. 7 illustrates a sensor circuit according to one embodiment;
- (9) FIG. 8 shows pulse and reflected waveforms according to one embodiment;
- (10) FIG. 9 is a functional diagram of a WMAS according to one embodiment;
- (11) FIG. 10 illustrates a moisture sensor circuit according to one embodiment;
- (12) FIG. 11 illustrates a series of wound moisture maps according to one embodiment; and
- (13) FIG. 12 illustrates a WMAS coupon integrated into different types of wound dressings.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

Overview

(14) In general, a Wound Moisture Assessment System (hereinafter ‘WMAS’) includes, inter alia, a flexible coupon configured to be placed on, over or about a wound such as, but not limited to, an ulcer. The coupon includes, inter alia, one or more moisture sensors configured to measure moisture in contact with, or in the immediate vicinity of the moisture sensor. The coupon includes requisite electronic circuitry to communicate with a WMAS wireless reader (hereinafter ‘reader’). The reader is configured to energize the coupon and receive measurements from the one or more moisture sensors. Moisture measurements can be analyzed using an analysis module integral with the reader, or, alternatively, moisture measurements can be transmitted to a computing system where analysis of wound moisture data can be processed for visualization, storage, integration with medical health records, transmission to a remote caregiver, or other functions. In a preferred embodiment, data transfer between the reader and coupon, and energization of the coupon by the reader occurs wirelessly. In general, the coupon can be configured with a desired number and type of moisture sensor(s), and a desired modality supporting wireless interrogation of the coupon by the reader.

(15) By integrating a plurality of moisture sensors into the coupon, a moisture ‘map’ of a wound bed can be obtained. These data can then be used by, for example, the patient or a caregiver to monitor the healing process of the wound or to alert the patient or caregiver of a critical moisture imbalance that may delay or preclude healing or indicate infection.

(16) For example, referring to FIG. 1, WMAS **100/200** (and use thereof) is illustrated according to one embodiment. As discussed in greater detail below, a WMAS can utilize an analog or digital moisture sensor data collection schema, which are discussed herein as WMAS **100** and WMAS **200**, respectively. Although coupon **201** is illustrated, FIG. 1 refers to coupon **101/201** and reader **120/220** to illustrate that, in a general sense, those elements are used in the same way by a patient, caregiver, doctor or other user. In this embodiment, the WMAS **100** includes a flexible coupon **101** that itself includes an array of moisture sensors.

(17) In this embodiment, the coupon **101** is configured to be applied over a wound, e.g., directly, in cooperation with an adhesive, or the coupon **101** can be integrated with a wound dressing as described in greater detail herein. The coupon **101** is configured for wireless communication with a wireless reader **120**. The coupon **101** can be interrogated (i.e., moisture sensor readings can be

obtained), by positioning the wireless reader **120** within range of the coupon **101** which triggers data collection of the moisture sensor(s). A single reading can provide real-time quantification of wound bed etiology; multiple sensor readings can be assembled into a moisture map to monitor wound healing over time or to identify that a wound is worsening or becoming infected, for example. Because interrogation of the coupon **101** is wireless, wound moisture and exudate levels can be determined without removing the protective coupon or dressing from the wound, which is one advantage of the WMAS **100** over traditional bandages and wound dressings.

(18) Interrogation of the coupon **101** by reader **120** can be accomplished by at least two approaches, each of which is described separately below. The first approach uses an analog radio frequency (RF) coupling between reader and coupon, and the second approach uses a digital schema. In both cases, the reader can communicate with a computing device **150**.

(19) Computing device **150** can be, without limitation, a personal computer, workstation, personal data assistant, cellular telephone, tablet or other electronic device that is capable of sending and receiving data signals and at least displaying wound moisture data as generally described herein. Computing device **150** can also communicate with reader **120** directly, or by known networking communications protocols, e.g., TCP, UDP, ICMP, HTTP, etc., through local (e.g., LAN), wide-area (e.g., WAN) networks, including the Internet, which can be accomplished by wired or wireless connections, e.g., Ethernet, LTE, WIFI, Bluetooth®, IR, WirelessHD, WiGig, etc., without limitation. Computing device **150** can be used, e.g., to display and store data collected by reader **120** in a user-friendly format, such as to display moisture maps as described herein.

(20) Reader/Coupon Wireless Coupling: Analog Schema

(21) Referring now to FIG. 2, a functional illustration of a WMAS **100** is shown according to one embodiment. In this embodiment, wireless coupling between the reader and coupon is accomplished using an analog schema. In the description that follows, the electronic circuit components of the coupon **101** can be disposed on coupon **101** by desired methods. However, a preferred method is ink-jet circuit deposition using a bio-friendly conductive ink such as silver nanoparticle ink.

(22) Referring to FIGS. 2-3, in this embodiment, the coupon **101** includes a flexible substrate **102** such a thin film of boPET (biaxially-oriented polyethylene terephthalate) having a top side (surface) **102a** and a bottom side (surface) **102b**. An antenna circuit (hereinafter ‘antenna’) **110** is disposed on the top side **102a** of the substrate and is configured for receiving radio frequency (RF) energy from a transceiver **125** of reader **120**. In this embodiment, at least one moisture sensor circuit **105** is disposed on the bottom side **102b** of the substrate. In this embodiment, each moisture sensor **105** is in electronic signal communication with coupon antenna **110**.

(23) In this embodiment, antenna **110** is one which can be activated by proximity-based inductive coupling with transceiver **125**, which provides a RF emission source. In general, the wireless reader **120** and the coupon **101** can communicate in a similar fashion as passive RFID systems. Each moisture sensor **105** is in electronic communication with the antenna **110**; e.g., they can be connected by a stenciled circuit path. This configuration allows RF energy received by the antenna **110** to be delivered to the at least one moisture sensor **105** for the purpose of obtaining a moisture value from each moisture sensor as described in greater detail herein. Such an approach advantageously eliminates the need for a separate moisture sensor power source, such as a battery or wired connection to the coupon **101**, providing the subject with the same type of unrestricted mobility as if he were wearing a regular wound dressing. Similarly, this configuration allows a coupon **101** to be placed on a wound for an extended period without needing to disturb the coupon to change batteries. This can reduce the likelihood of infection and minimize disturbance to the wound bed itself.

(24) In this embodiment, transceiver **125** is configured to emit a controllable activation signal, e.g., an RF signal that is sufficient to power the at least one moisture sensor **105** when a moisture sensor reading is desired. In a preferred embodiment, antenna **110** is configured to receive the RF

activation signal from transceiver **125** with minimal energy loss and shunt the energy to the at least one moisture sensor **105**. In this embodiment, each of the one or more moisture sensors **105** can be used to collect a moisture sensor reading, e.g., a moisture value, or other value that can be converted or interpreted as a moisture level, which can be communicated back to transceiver **125** via antenna **110** as described in greater detail herein.

(25) In coupon embodiments where multiple moisture sensors are used, each moisture sensor **105** can optionally have a separate, dedicated antenna operably associated with it. In such an embodiment, each dedicated antenna can encompass the associated moisture sensor so that each moisture sensor/antenna pair can be interrogated individually by the reader **120**. In a preferred approach, a programmable microcontroller **130** (discussed below) can be configured to activate and collect a moisture reading from each moisture sensor **105** individually, e.g., in a round-robin fashion.

(26) Alternatively, in a different embodiment, a single antenna can energize an array of moisture sensor circuits, wherein each moisture sensor circuit of the array can be configured with a unique conductance and capacitance to provide a uniquely-identifiable sensor impedance range. In such an approach, the backscattered resonant waveform can then include several resonant peaks, e.g., a unique peak frequency for each moisture sensor circuit, wherein each moisture sensor circuit can be present in a specific frequency band. The frequency band can be dependent on the characteristic impedance of each moisture sensor circuit.

(27) In this embodiment, the reader **120** includes a programmable microcontroller **130** in electronic signal communication with the transceiver **125**, a communications module **135** and a user interface module **140**. In a preferred embodiment, microcontroller **130** includes at least a processor, a memory, and electronic data and command storage capabilities, e.g., RAM, ROM, a disk drive, cache, or other module for electronically storing instructions for activating the one or more moisture sensors **105** and receiving moisture sensor readings therefrom, preferably analyzing those data, and transmitting the moisture sensor reading results to a remote computing device.

(28) In this embodiment, microcontroller **130** is in signal communication, e.g., through the use of an input/output port with user interface **140**. User interface **140** can include, without limitation, a screen for displaying information relating to the acquisition of moisture sensor data, moisture analysis results and related data, historical data, and other information. In a preferred embodiment, user interface **140** can include a touch-sensitive display screen that provides the capability of entering user input, e.g., patient-identifying information, general system settings (date, time, interrogation frequency, etc.), computer network settings providing the capability of transmitting data from the WMAS **100** to a computer network, e.g., through the use of wired or wireless signal communications, activation buttons configured to initiate a wound moisture assessment, and other functions.

(29) In this embodiment, microcontroller **130** is in signal communication with a communications module **135** that is configured to transmit moisture sensor readings to a remote computing device **150**. The communications module **135** can be configured to provide bi-directional signal communication between the reader **120** and the remote computing device **150** by any preferred wired or wireless communications protocol or standard, such as WIFI, BLUETOOTH, IR, etc. In this embodiment, for wireless signal transmission, the communications module **135** communicates with remote computing device **150** using wireless antennas **145** and **155**, respectively. In one embodiment, the wireless reader can be in the form of an RFID scanner gun which is configured to collect moisture sensor readings and transmit those data to computing device **150** for processing into a user-friendly format such as wound moisture map. In another embodiment which can also be applicable to WMAS **200**, the reader can be integral with a smartphone or other personal communications device equipped with near-field scanning capabilities. In such an embodiment, the smartphone can include a software interface for activating scanning functionality and sending, storing and displaying data associated with a moisture sensor scan.

(30) In this embodiment, to interrogate coupon **101** for the purpose of receiving a wound moisture assessment, the microcontroller **130** is configured in part to cause transceiver **125** to emit a pulse-width modulated (PWM) activation signal which can either directly drive the antenna **110**, or, alternatively, the activation signal can be fed through an adjustable low-pass filter to supply a sine wave (SW) signal to the antenna **110**. In this embodiment, either the PWM or SW signal is configured to generate an electromagnetic field coupling between the antenna **110** of the coupon **101** and the transceiver **125** of the wireless reader **120**. In this embodiment, the microcontroller can generate a regulated alternating-current (AC) voltage signal that is applied to the transceiver **125** while simultaneously measuring the corresponding impedance of the at least one moisture sensor tag **105**, as AC frequency f is varied. In this way, in this embodiment, moisture sensor readings are communicated from the coupon **101** to the reader **120**.

(31) Referring to FIG. 4, a moisture sensor circuit e.g., of moisture sensor **105** is illustrated according to one embodiment. In this embodiment, moisture sensor **105** includes in part a resistor R , inductor (coil antenna) L , and capacitor C in a parallel (FIG. 4, A) or series (FIG. 4, B) resonant circuit configuration, also known as an RLC circuit as described in greater detail below. In this and other embodiments, the capacitor element of the RLC moisture sensor circuit can operate as an electrochemical transducer to measure or determine wound moisture.

(32) Referring to FIG. 5, an exemplary circular loop air coil antenna, such as antenna **110** of reader **120**, is shown in top (A) and side (B) views. The coil antenna can be a multi-layer coil antenna to form an inductance coil in limited space for wirelessly interrogating the one or more moisture sensors **105** of coupon **101**. The at least one sensor **105** can be wirelessly interrogated by an electromagnetic field whose parameters depend at least in part on the physical design of the antenna, namely: an electric current I passing through the antenna coil, the antenna average radius r , the number of turns N in the antenna coil, the read-distance x along the central axis of the coil, the coil winding thickness t and coil winding height h . In this embodiment, the electromagnetic field created by antenna **110** induces a current and a potential drop in the at least one moisture sensor **105** of coupon **101**.

(33) In this embodiment, the antenna coil includes N turns, where r is the average radius of the coil, t is the winding thickness and h is the winding height. Using the inductance formula:

$$(34) L = \frac{0.31(rN)^2}{6r + 9h + 10t}$$

with $N=200$, $r=1.0$ inch, $h=0.05$ cm (19.68 mils) and $t=0.5$ cm as preferred values according to one embodiment, a value of $L=3.87$ mH is obtained. To form a resonant circuit for 125 kHz RF operation, a resonant capacitor value can be calculated by the expression:

$$(35) C = \frac{1}{(2\pi f)^2 L}$$

where f is the frequency (125 kHz) and L is 3.87 mH; a resonant capacitance value of 419 pF is thus computed in this exemplary embodiment.

(36) Referring now to FIG. 6, in general, each moisture sensor circuit can possess a characteristic (or resonant) frequency f and bandwidth B . Without wishing to be bound by theory, it has been discovered that the bandwidth, resonant frequency, and complex impedance Z of the RLC circuit of sensor **105** (FIG. 6A) is dependent upon a capacitance value, which itself is dependent upon a moisture level of the environment surrounding the at least one sensor **105**. Thus, referring to FIG. 6B, a change in the moisture level of the surrounding environment can cause a corresponding, measurable shift and damping of the resonant frequency of the RLC resonator.

(37) Still referring to FIG. 6B, in this example, the solid line shows sensor resonant impedance as a function of frequency. The bandwidth B of the sensor **105** circuit can be defined as the range between the peak of the maximum and minimum resonant values. The point corresponding to the lower frequency at half-power is $f_{sub.L}$ and referred to as the lower cut-off frequency with the point corresponding to the upper cut-off frequency as $f_{sub.H}$. The range between $f_{sub.H}$ and $f_{sub.L}$ can represent the bandwidth B as illustrated. The dashed line illustrates a shift and damping

of the resonant impedance as a function of frequency (i.e., bandwidth) for a sensor **105** in a moist environment, where the complex impedance Z is given by the range between minimum and maximum values. Such a shift can be detected and analyzed, e.g., by microcontroller **130** of reader **120** as an impedance reflected in the antenna **110**. This phenomenon can be exploited for qualifying and quantifying moisture levels of a wound bed when the sensor **105** is placed on or adjacent thereto.

(38) Referring now to FIG. 7, in this embodiment, each moisture sensor **105** is configured as an RLC resonator circuit. In this embodiment, the moisture sensor **105** includes a four-element electrode array (electrodes **107a-d**) that together form a moisture transducer **106** and which acts as the capacitor in the RLC circuit. In this embodiment, each electrode (e.g., each of electrodes **107a-d**) has a height dimension of about 1/8-inch, a width dimension of about 1/16-inch and the electrode-to-electrode spacing is about 1/32-inch. While the electrode dimensions disclosed here are suitable to enable the instant embodiment, it should be understood that other electrode dimensions and configurations can be substituted according to preference or to achieve operational characteristics as desired.

(39) In this embodiment, the RLC circuit of the moisture sensor **105** further includes a substantially C-shaped resistor element **108** and a substantially ring-shaped inductor element **109** that surrounds both the C-shaped resistor element **108** and moisture transducer **106** as illustrated. Such a circuit can be printed, e.g., on a polyester film having about a five millimeter thickness with silver conductive polymer ink such that the moisture transducer **106**, resistor **108** and inductor element **109** have thicknesses of about 1 mm. Exemplary printable conductors suitable for use in this capacity are DuPont's **5007** or **5025** printable silver conductors sold by DuPont, Research Triangle Park, North Carolina, USA; screen-printable silver ink nos. **1210** and **6105** provided by Methode (Chicago, Illinois, USA); and product no. ECI **1005** provided by Henkel Electrical Materials, Irvine, California, USA.

(40) Referring back to FIG. 3, in this and other embodiments, coupon **101** can include a plurality of moisture sensors **105** arranged as desired to achieve a preferred moisture sensor density across coupon **101**. In FIG. 3, each individual moisture sensor is labeled **S1-S8** to correspond with the discussion of moisture mapping that follows. In the exemplary illustration shown in FIG. 3, a plurality of moisture sensors **105** are in electronic communication with (i.e., through circuit connections), the centrally-disposed antenna **110** as shown. In this embodiment, each moisture sensor **105** of the coupon **101** can be configured to be individually readable by reader **120** for collecting wound moisture and exudate levels in an area adjacent to, or in contact with each individual sensor **105**.

(41) To achieve individual moisture sensor readings, in one embodiment, each sensor **105** can be configured to have a unique, identifiable resonant impedance profile, e.g., a resonance pattern through control of each printed sensor circuit. In such an embodiment, each resonant impedance profile can be measured for each sensor of the plurality of sensors and correlatively stored in memory. Each sensor can be identified by, e.g., one or more of a unique resonant impedance waveform shape (e.g., the solid line illustrated in FIG. 6B), unique positions or values of inflection points of the resonant impedance waveform, the amplitudes of selected resonant impedance waveform attributes, or a shift along the frequency axis—e.g., a change in bandwidth—of the impedance waveform. In general, the resonant impedance waveform can be measured for each individual sensor and correlatively stored with a particular sensor, e.g., sensor **S1**, sensor **S2**, etc. In one embodiment, each sensor **105** disposed on coupon **101** can be individually interrogated for a surrounding moisture level reading by modifying the transceiver output to match the characteristic resonant frequency of any selected sensor **105**. In one embodiment, microcontroller **130** can be configured to make such output modifications in a step-wise order to interrogate each sensor present on a coupon **101**.

(42) In this and other embodiments, practices used in the RFID industry for minimizing signal

collisions between tags can be applied to coupons having a plurality of sensors. For example, a WMAS **100** can be configured as a 'slotted Aloha' system where the activation of each sensor is delayed in a particular order, e.g., clockwise beginning with a sensor at the 12 o'clock position. In another approach, the WMAS **100** can be configured to utilize a so-called 'adaptive binary tree' protocol, where each sensor **105** is configured with a pre-programmed ID bit and is activated only when the wireless reader **120** transmits an initialization symbol and a matching bit ID or ID sequence. In yet another approach, an integrated circuit can be disposed on the coupon **101** and configured such that activation energy transmitted by the reader **120** and received by antenna **110** is shunted to each sensor **S1-S8** in a particular order. For example, the integrated circuit can be configured to intercept the activation energy received by antenna **110**, then shunt the activation energy to a first sensor **S1**, followed by a second, different sensor **S2**, and so on, until all sensors **S1-S8** have been activated in a selected order. Each sensor reading can then be collectively used to create a moisture map.

(43) Still referring to FIG. **3**, in this embodiment, coupon **101** includes eight moisture sensors **105**, labeled **S1-S8**; however, it should be understood that more or fewer moisture sensors can be used as preferred. The array of moisture sensors can be used for creating a map of wound moisture or exudate. In a preferred approach, such a map can be collected at selected intervals over a time period to monitor and manage acute and chronic wounds.

(44) Referring now to FIG. **8**, in this and other embodiments, determination of moisture or exudate levels and content from each moisture sensor can be accomplished, for example, using pulse symmetry analysis, alone or in combination with other analytical approaches. As described above, in this embodiment, moisture sensor **105** readings of coupon **101** can be obtained when reader **120** is activated to emit a PWM signal. The PWM signal can be received by antenna **110**, and thereby generate an electromagnetic field coupling between the transceiver **125** and antenna **110** of the coupon **101**. Microcontroller **130** can be configured to generate a regulated AC voltage signal that is applied to the transceiver **125** and, simultaneously, measure the corresponding impedance of the at least one coupon sensor **105** as AC frequency is varied. The resulting waveform, which correlates impedance as a function of AC frequency (hereinafter referred to as an "I/F waveform"), includes at least three features from which a moisture determination can be made: 1) the overall shape of the I/F waveform; 2) the amplitude of the I/F waveform; and 3) the symmetry of the I/F waveform pulse, which in this approach reflects the balance between leading and falling edges of the I/F waveform.

(45) Without wishing to be bound by theory, it has been discovered that the amplitude, or overall height of the I/F waveform, e.g., relative to zero amplitude, can be used to reliably indicate the amount of moisture present in the vicinity of a sensor **105**, and wet/dry cycling. The shape and symmetry of a I/F waveform can furthermore provide detailed information on the properties and chemistry of the fluid environment, such as electrochemical impedance, pH level and chemical composition (e.g., electrolyte identification), among other information.

(46) In this embodiment, the reader **120** can be configured to power and communicate with the coupon **101** using a time pulse method. In this method, the reader **120** initiates a scan to read coupon **101** moisture by generating a time series pulse of known amplitude, width, and duty cycle. The antenna **110** captures the RF pulse wave transmitted from the reader **120** and the reflected impedance of the moisture sensor **105** is then analyzed by a time response analyzer (TRA) module **133** in the reader **120**. The TRA module **133** can be an electronic circuit module configured to quantify the pulse symmetry of the reflected I/F waveform from the sensor **105**, for example: 1) the overall shape of the I/F waveform, 2) the amplitude of the I/F waveform, and 3) the symmetry of the I/F pulse, which, in this approach can represent a balance between the leading and falling edges of the reflected I/F waveform.

(47) For example, FIG. **8** shows scan pulse and reflected I/F waveforms (charts A-D) of WMAS **100** test data for a coupon **101** exposed to varying degrees of moisture. In this example, each set

shows the reader **120** scan pulse waveform in the top chart, e.g., chart **A1** and the reflected, or output signal I/F waveform from a sensor **105** in the bottom chart, e.g., in chart **A2**. Referring to chart **A**, in this example, a coupon **101** was immersed in a tap water solution to represent a soaked wound bandage. As compared to the scan pulse **A1**, in this example, the reflected I/F waveform **A2** has a decreased amplitude but its bandwidth is substantially narrow, similar in magnitude to the scan pulse **A1**.

(48) Referring to chart **B** of FIG. **8**, in this example, the coupon was exposed to a damp environment by applying two drops (1.7×10^{-3} fluid ounces per drop) of tap water solution to a coupon **101**. The resulting output I/F waveform **B2** is broadened, e.g., measured at its full-width-half-maximum (FWHM) and shows a decreased amplitude compared to the scan pulse **B1**. Charts **C** and **D** show the effect of the I/F waveform shape, e.g., amplitude and width at FWHM showing a progressive shift toward greater output signal amplitude with increasing dampness and, concurrently, a greater negative deviation or 'recovery' following the excitation pulse with increasing sensor dampness.

(49) In one embodiment, a correlative look-up table or a predictive mathematical model can be constructed from standard samples such as those shown in the charts of FIG. **8**, so that dampness and other factors of a wound bandage can be accurately determined by the WMAS **100**. For example, as FIG. **8** illustrates, and without wishing to be bound by theory, the amplitude of an I/F waveform appears to be dependent at least in part on the degree of dampness of the coupon **101** (and sensor **105** environment). Thus, the shape of the I/F waveform in general, e.g., amplitude, inflection and deflection points and areas, pulse width and other factors can be used to qualify and quantify exudate levels on or near a wound bed as described herein.

(50) Reader/Coupon Wireless Coupling: Digital Schema

(51) Referring now to FIG. **9**, a functional illustration of a WMAS **200** is shown according to one embodiment. In this embodiment, wireless communication between the reader and coupon can be accomplished using a digital circuitry schema. For the sake of consistency throughout this disclosure, in the description that follows, reference to coupon, moisture sensor(s), reader and computing device refer to the WMAS components in a general sense as previously described, notwithstanding the differences of the analog and digital components of the reader and coupon in each respective section of this disclosure.

(52) In this embodiment, the coupon **201** and reader **220** of WMAS **200** are configured with circuitry and modules for supporting digital acquisition of moisture sensor readings and digital transmission of moisture readings and associated data from the coupon **201** to the reader **220**. Like the analog WMAS **100** counterparts, some circuit components, including, but not limited to the moisture sensors **205** can be disposed on coupon **201** by a selected circuit-printing technique. For example, copper etching, a technique commonly used in the production of RFID tags can be used. To facilitate rapid and inexpensive circuit deposition, however, the technique of inkjet circuit printing can be used. In this preferred technique, the electronic circuitry of coupon **201** can be printed directly on the coupon substrate using, preferably, a bio-compatible ink such as a silver nanoparticle ink. Other integrated circuit elements, such as control module **170** described below can be adhered to, or integrated into the coupon substrate by known methods.

(53) In this embodiment, coupon **201** includes a control module **170** which, in this embodiment can be a fully-operational system-on-a-chip (SoC) configured for near-field communications (NFC) between the coupon **201** and reader **220**. The control module **170** can function, in part, as a transceiver, similar to transceiver **125** discussed with respect to FIG. **2**. In general, similar to the analog counterpart, the circuitry of coupon **201** can be powered by energy transmitted from an antenna **120a** of the reader **220**. For example, the coupon **201** can include a power module **169** configured to receive RF energy which can be in circuit communication with other components of the coupon **201** requiring energy to function. The coupon **201** can thusly be energized to engender one or more data measurements to be taken, such as one or more moisture level readings as

described in greater detail below. Once the measurements are complete, the measurement data can be transmitted by coupon **201** to reader **220**. For example, the control module **170** can include a near-field communications module configured to transmit data from the coupon **201** to the reader **220**. In one embodiment, the reader **220** can supply energy to the control module **170** through electromagnetic coupling and the digital response from the coupon **201** can be transmitted through backscatter to the reader **220**. In a preferred embodiment, this form of energy harvesting and data transfer is accomplished through electromagnetic induction between the reader **220** and the coupon **201**. In this embodiment, reader **220** can be configured in a similar fashion as a radio frequency identification (RFID) scanner.

(54) In this embodiment, control module **170** includes an ISO15693 radio-frequency interface with coupon antenna **175**. In this embodiment, antenna **175** is a printed circuit disposed on a top side **102a** of the coupon substrate **102** that is configured to harness energy transmitted by reader antenna **120a** necessary to power circuitry, modules and any other components of the coupon **201** through inductive coupling. Accordingly, in this embodiment, the antenna **175** circuit is in electronic communication with at least the control module **170** and the power module **169**. As such, coupon **201** does not necessarily require an on-board, stored power source such as a battery or wired electrical connection to carry out the functions of the coupon **201** as described herein, although such a configuration could be used in an alternative embodiment. The wireless configuration described herein advantageously provides the wearer with the same type of unrestricted mobility as if he were wearing a regular wound dressing.

(55) In this embodiment, control module **170** can be a stand-alone near-field communications sensor transponder such as the RF430FRL152H sensor transponder by Texas Instruments, Dallas, TX, USA. In such an embodiment, the sensor transponder can be in circuit communication with one or more moisture sensors **205**, temperature sensor **173**, and bacteria sensor **182** of the coupon **201**.

(56) In general, the control module **170** can store and analyze moisture sensor data to interpret results, e.g., moisture map data (as described in greater detail below) can be tabulated and transmitted to the reader to be displayed for clinical interpretation. Alternatively, or in combination, moisture sensor data can be stored and analyzed in the controller **130** of the reader **220**. In one approach, a shared processing routine can utilize both controllers **171**, **130**; e.g., the coupon processor **170** can perform moisture sensor data preprocessing, filtering and formatting and the reader controller **130** can perform display and post measurement signal processing, including linearizing moisture sensor data.

(57) In one embodiment, control module **170** can include an integrated circuit microcontroller **171**. Microcontroller **171** can be selected from available commercial sources according to preference to address desired performance characteristics, power consumption or other aspects of coupon **101** functionality. One exemplary, non-limiting microcontroller that can be used is the MSP430 ultra-low power, 16-bit microcontroller provided by Texas Instruments, Inc., Dallas, TX, USA. In this embodiment, microcontroller **171** is in electronic communication with the modules, sub-modules and sensor circuitry of the coupon **201**, and is programmable to collect measurements, e.g., moisture and bacteria sensor measurements, alone or concurrently, as described herein. Microcontroller **171** is also in signal communication with antenna **175**, enabling sensor measurements to be transmitted to reader **220**.

(58) In this embodiment, control module **170** further includes a conversion sub-module **172** to convert digital data, e.g., moisture sensor data as described herein, to wireless signals according to a NFC standard codec.

(59) In this embodiment, control module **170** further includes first (**176**) and second (**177**) memory modules that can be used, e.g., for storing executable software instructions for carrying out various data-collecting functions as described herein, such as collecting sensor measurements, and storing data associated with those measurements. In one example, the first memory module can be a 2 KB

non-volatile ferroelectric memory, and the second memory module can be an 8 KB read-only memory (ROM).

(60) In this embodiment, the control module **170** can optionally include an electronic temperature sensor. The temperature sensor can be used, e.g., to linearize moisture sensor data. In embodiments that include a temperature sensor, temperature measurements can be recorded and transmitted along with moisture sensor measurements. In this and other embodiments, temperature and moisture sensor data can be cooperatively used to correlate with bacteria sensor **182** data to assess the impact of bacteria on moisture sensor readings and to model bacterial activity.

(61) In this embodiment, coupon **201** further includes an analog front-end component **184**. In this embodiment, the analog front-end component includes a multiplexer **186** that is configured to allow controller **170** to excite and gather moisture measurements from one or more moisture sensor circuits **205** of the coupon. In this and other embodiments, coupon **201** can include as many moisture sensor circuits as desired. For example, in this embodiment, moisture sensor array **180** includes a plurality of individual moisture sensors **205** which are each configured to operate as electrochemical transducers to measure wound moisture. In general, controller **170** can be configured to excite and gather data for all moisture sensors **205** concurrently, individually or according to a predetermined pattern.

(62) Referring to FIG. **10**, digital acquisition of moisture sensor **205** data in this embodiment differs from the previously-described analog counterpart mainly in that each sensor **205** is configured as a stand-alone electrochemical transducer that is in signal communication with microcontroller **171** via analog multiplexer **186**. In this embodiment, like the analog counterpart, each sensor **205** includes interdigitated electrodes **107a-d**. However, direct multiplexing obviates the resistor R and inductor L sensor circuit elements used in the analog counterpart (see, e.g., FIG. **7**). Thus, in the digital schema presently discussed, capacitance is used to model an impedance change which is correlatable to a moisture level across sensor electrodes **107a-d**. In the analog counterpart, the inductor L is used to model the transmitter and receiver antenna coupling, which through the waveform analysis discussed previously, can provide correlation to moisture levels across electrodes **107a-d** (FIG. **7**).

(63) In this embodiment, the analog front-end component **184** further includes an operational amplifier **188** as an interface to an electrochemical bacteria sensor **182**. Bacteria sensor **182** can be used, e.g., to detect and quantify contamination of moisture sensors **205** from exudate due to proliferation of bacteria and onset of infection. In this embodiment, bacteria sensor **182** is centrally disposed on bottom side **102a** of the coupon substrate and includes a Ag/AgCl reference electrode **182a**, and a carbon-based working (**182b**) and counter (**182c**) electrode.

(64) In one approach, the influence of exudate on sensor **205** impedance can be quantified and corrected for by, e.g., modeling bacterial contamination. Without wishing to be bound by theory, it is believed that impedance ‘interference’ due to bacterial contamination is caused by two primary sources: the first being microbial metabolism, which alters the conductivity of the medium; the second is electrode interfacial impedance, results when the presence of bacteria changes the surface properties of the electrodes, which can affect the capacitance of the electrode/electrolyte interface. Thus, growth of microorganisms such as bacteria usually results in an increase in both conductance and capacitance, causing a decrease in impedance. In this embodiment, temperature sensor **173** can provide an additional modeling and measuring parameter to account for the temperature effect on electrode capacitance, metabolic behavior of bacteria, and evaporation

(65) Referring back to FIG. **9**, in this embodiment, wireless reader **220** includes a microcontroller **130** in electronic signal communication with antenna **120a**, a communications module **135** and a user interface **140** which itself is in signal communication with an external display **136**. In a preferred embodiment, display **136** is a touch-screen display device that allows a user to input information, control functionality of the reader, activate moisture sensor scans, etc. However, reader **220** can optionally include buttons, keyboards or other user-controllable input devices which

are not shown in FIG. 9 for the sake of figure clarity.

(66) In this embodiment, controller **130** is configured to emit a controllable activation signal, e.g., an RF signal from antenna **120a** that provides sufficient energy to power energy-necessary components of coupon **201**, e.g., controller **170**, and to allow moisture sensor **205** measurements to be taken. In a preferred embodiment, antenna **175** of coupon **201** is configured to receive the activation signal from reader **220** with minimal energy loss. In this embodiment, each of the one or more moisture sensors **205** can be used to collect a moisture sensor reading which is then transmitted back to reader **220** via antenna **175**.

(67) In a preferred embodiment, microcontroller **130** includes at least a processor, a memory, and electronic data and command storage capabilities, e.g., RAM or ROM for electronically storing instructions for activating the one or more moisture sensors **205** and receiving moisture sensor readings therefrom, preferably analyzing those data, and transmitting the moisture sensor reading results to the computing device **150**.

(68) In this embodiment, microcontroller **130** is in signal communication, e.g., through the use of an input/output port, with user interface **140**. User interface **140** can include, without limitation, a screen for displaying information relating to the acquisition of moisture sensor data, moisture analysis results and related data, historical data, and other information. In a preferred embodiment, user interface **140** can include a touch-sensitive display screen **136** that provides the capability of entering user input, e.g., patient-identifying information, general system settings (date, time, interrogation frequency, etc.), computer network settings providing the capability of transmitting data from the WMAS **100** to a computer network, e.g., through the use of wired or wireless signal communications, activation buttons configured to initiate a wound moisture assessment, and other functions.

(69) In this embodiment, microcontroller **130** is in signal communication with a communications module **135** that is configured to transmit moisture sensor readings to computing device **150**. The communications module **135** can be configured to provide bi-directional, electronic communication between the WMAS **100** and the remote computing device **150** by any preferred wired or wireless communications protocol or standard, such as WIFI, BLUETOOTH, IR, etc. In this embodiment, for reliable signal transmission, the communications module **135** communicates with remote computing device **150** using wireless antennas **145** and **155**, respectively. In a preferred embodiment, the wireless reader **220** can be in the form of an RFID scanner gun which collects moisture sensor readings and transmits those data to computing device **150** so that they may be processed into a user-friendly format such as a wound moisture map.

(70) In this embodiment, coupon **201** can be powered through electromagnetic induction with reader **220** RF emission at, e.g., NFC frequencies of 13.56 MHz or other chosen frequencies. The reader **220** can use NFC standard techniques to power and communicate with the coupon; in such a case, the reader **220** can be implemented with a smartphone or other standard NFC device.

(71) In this and other embodiments, reader **220** can be configured to transmit moisture sensor measurements from coupon **101** to computing device **150**. Communication between the reader **220** and computing device **150** can be accomplished by any preferred wired or wireless communications protocol or standard, such as Ethernet, WIFI, BLUETOOTH, IR, etc. In general, moisture sensor measurements need not be directly between the reader **220** and the computing device **150**. For example, reader **220** can be in signal communication with a wireless router, access point or other communications hub which directs the measurement traffic to computing device **150** via a computer network, such as a LAN or the Internet, by known methods as disclosed above.

(72) In this and other embodiments, computing device **150** can be configured to display information relating to the acquisition of moisture sensor data, moisture analysis results and related data, historical data, and other information via a software package. In a preferred embodiment, reader **220** can be configured to allow the user to enter information, such as the user's name, a patient ID, location, bed number, wound location, date, time, wound interrogation frequency or

other data through a programmable user interface. The user-supplied information can be transmitted along with moisture sensor measurements from the reader **220** to the computing device **150** and, for example, compiled to form a data set. In one embodiment, the user interface can include a touch-sensitive display screen that provides the capability of entering user input, computer network settings providing the capability of transmitting data from the reader **220** to a computer network, e.g., through the use of wired or wireless signal communications, activation buttons configured to initiate a wound moisture assessment, and other functions.

(73) Moisture Mapping

(74) Referring now to FIG. **11**, in general, WMAS **100** can be utilized to create a wound moisture map **160** of a wound. A moisture map of the type described herein can reflect moisture sensor data whether obtained by the analog or digital schemas described herein. A wound moisture map **160** can be a graphical representation of the moisture levels of a wound, as measured in different areas of the wound by the sensors **105/205** of coupon **101/201**. In general, a wound moisture map **160** can help to ascertain wound moisture imbalance and, in a preferred approach, such a map can be collected at selected intervals over a time period to monitor and manage acute and chronic wounds. (75) For example, FIG. **11** shows a plurality of moisture maps **160** for first, second, third and fourth dressings taken at 4, 8, 16 and 24-hour time intervals. In this illustration, each labeled square **S1-S8** represents a moisture level measured by a corresponding moisture sensor **S1-S8** of coupon **101** illustrated in FIG. **7**. For example, square **161** graphically depicts the measured moisture level of the wound corresponding to sensor **S6** of coupon **101**.

(76) In this and other embodiments, moisture map **160** can be graphically displayed using a legend to indicate a qualitative or quantitative moisture sensor reading. In the examples shown in FIG. **11**, each moisture map **160** graphically depicts moisture sensor readings according to a “Dry,” “Slightly Wet,” “Semi-Wet” and “Wet” legend scale. The legend scale can be selected according to preference to correlate to a quantitative moisture level as measured by the moisture sensors **105**. Alternatively, moisture map **160** could display quantitative data that show actual moisture levels as measured by each sensor **105/205**, which can be expressed in any desired measurement unit.

(77) In one approach, a coupon **101/201** provides the ability to dispose one or more moisture sensors at the perimeter of a wound where moisture sensing can be critical to: 1) understand where undermining (e.g., deep tissue damage under the wound margin) is occurring; 2) interpreting where maceration exists or where the locations where the wound perimeter is too dry, which can limit blood flow and irrigation of the wound bed; 3) ascertaining the true border of the wound; 4) ascertaining where new tissue could be forming or where granulation of tissue (epithelialization) is occurring; and 5) ascertaining where, and if infection is present.

(78) FIG. **11** shows an array of moisture maps **160**, each for a first through fourth dressing. For example, the top-left moisture map **160** corresponds to a measurement of a first coupon **101** (which may, e.g., be integrated into a wound dressing) at the 4-hour interval, and indicates that the respective portions of a wound adjacent to sensors **S1-S8** are dry. At the 8-hour interval, the reading for moisture sensor **S6** indicates a slightly wet environment; at the 16-hour interval, the reading for moisture sensor **S6** continues to indicate a slightly wet environment and moisture sensor **S4** indicates a semi-wet environment; at the 24-hour interval, moisture sensor **S6** indicates a dry environment while sensor **S7** indicates a slightly wet environment, and sensors **S4** and **S3** indicate semi-wet and wet environments, respectively.

(79) The exemplary data shown in FIG. **11** illustrate how coupon **101/201** having an array of moisture sensors **105/205** can be used to monitor physiological changes in a wound bed over time; accordingly, these data can be used in caring for the wound or providing an alert that wound healing is occurring or not. The beneficial effects of a moist versus dry wound environment include: prevention of tissue dehydration and cell death, accelerated angiogenesis, increased breakdown of dead tissue and fibrin, e.g., pericapillary fibrin cuffs, and potentiating the interaction of growth factors with their target cells. In this and other embodiments, the extrapolation of

moisture sensor data (explained in greater detail below) or the sensors themselves can be calibrated according to preference so that the relative terms “wet” “slightly wet” “semi-wet” and “wet” can reflect different degrees of wound moisture or exudate in a moisture map. In a complex wound such as a pressure ulcer being treated with a vacuum-assisted closure (VAC) device, for example, points of desiccation and alternatively areas of pooling of exudate may exist. The WMAS **100/200** can identify these areas and allow caregivers or the patient himself to modify their care to optimize the moisture of the wound for proper healing.

(80) Referring now to FIG. **12**, coupon **101/201** can be integrated into a protective wound dressing or bandage of any kind. For example, coupon **101/201** can be disposed on one side of a protective dressing and configured such that it can be applied directly to a wound, such as, but not limited to a venous, arterial, pressure or diabetic wound. In an alternative embodiment, the coupon **101/201** can be integrated into a protective dressing such that it is disposed between top and bottom layers of the dressing. In such an approach, the dressing can provide at least one layer of protective material between the wound bed and the coupon **101/201**.

(81) Some exemplary, non-limiting types of dressings that coupon **101/201** can be integrated with include: tape and foam dressings, where coupon **101/201** can be disposed between the foam layer and the cover layer; transparent films, where coupon **101/201** can be printed directly on the film window; gauze, where coupon **101/201** can be embedded therein or thereon; composites, where coupon **101/201** can be embedded in intermediary layer or printed on the cover layer; and alginate, where coupon **101/201** can be disposed therein or on a secondary dressing.

(82) In one embodiment, an exudate anti-fouling layer can be integrated into the dressings to reduce the deleterious effects of exudate adherence and acidity build-up between the wound and the dressing. This is illustrated in FIG. **12** by the dashed line between the intermediary layer and the absorbent layer. In such an embodiment, the anti-fouling layer can be perforated to promote exudate diffusion therethrough; and, in one embodiment, the holes of the perforated layer can contain exudate filter material to minimize exudate contamination of the coupon **101/201** components. Other applicable dressings will be evident to those skilled in the art.

(83) A number of illustrative embodiments have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the various embodiments presented herein. For example, other moisture sensor configurations and layouts can be substituted for those described herein. At least one multi-element sensor array is described in U.S. patent application Ser. No. 14/690,324 filed on Apr. 17, 2015 by J. N. Schoess et al., which is incorporated by reference in its entirety herein. Accordingly, other embodiments are within the scope of the following claims.

Claims

1. A coupon for wound care adapted for wireless communication with a transceiver, comprising: a perforated adhesive layer adapted to secure the coupon to skin; a flexible substrate coupled to the perforated adhesive layer and having a top surface and a bottom surface; a radio-frequency antenna circuit disposed on the top surface of the flexible substrate, the radio-frequency antenna circuit being adapted to receive radio frequency energy; a plurality of electrochemical moisture sensor circuits disposed on the bottom surface of the flexible substrate, each configured to be powered by the radio frequency energy; and a microcontroller in signal communication with the radio-frequency antenna circuit and the plurality of electrochemical moisture sensor circuits, the microcontroller adapted to: obtain data from each electrochemical moisture sensor circuit of the plurality of electrochemical moisture sensor circuits; and generate, based on the data from the plurality of electrochemical moisture sensor circuits, an operating state of the coupon from a set of operating states, the set of operating states comprising: a first operating state that indicates a first moisture level between a first minimum value and a first maximum value; and a second operating

state that indicates a second moisture level between a second minimum value and a second maximum value that are each greater than the first maximum value.

2. The coupon of claim 1, further comprising: a near-field communications module disposed on the top surface of the flexible substrate that is in electronic communication with the radio-frequency antenna circuit.
3. The coupon of claim 1, further comprising: a memory for storing executable logic functions that when executed by the microcontroller causes the microcontroller to measure a moisture value of an environment proximal to the at least one electrochemical moisture sensor circuit.
4. The coupon of claim 1, further comprising: a multiplexer in signal communication with the microcontroller and the plurality of electrochemical moisture sensor circuits; a signal converter capable of converting analog signals to digital signals and vice-versa; and a temperature sensor.
5. The coupon of claim 2, further comprising: a bacteria sensor disposed on the bottom surface of the flexible substrate that is in signal communication with the near-field communications module.
6. The coupon of claim 2, wherein the near-field communications module is configured to receive operational power from an external radio-frequency source that is received by the radio-frequency antenna circuit.
7. A system for wound care adapted for wirelessly obtaining a moisture measurement, comprising: a wireless reader comprising a transceiver configured to emit and receive radio frequency signals; and a coupon, comprising: an adhesive layer adapted to secure the coupon to skin; a flexible substrate coupled to the adhesive layer and having a top surface and a bottom surface; at least one electrochemical moisture sensor circuit disposed on the bottom surface of the flexible substrate that is configured to determine a moisture value; an antenna circuit disposed on the top surface of the flexible substrate configured to at least receive the radio frequency signals from the transceiver and to power the at least one electrochemical moisture sensor circuit by radio frequency energy of the received radio frequency signals; and a microcontroller in electronic communication with the antenna circuit and the at least one electrochemical moisture sensor circuit, the microcontroller adapted to: obtain data from the at least one electrochemical moisture sensor circuit; generate, based on the data, an operating state for the coupon from a set of operating states, the set of operating states comprising: a first operating state that indicates a first moisture level between a first minimum value and a first maximum value; and a second operating state that indicates a second moisture level between a second minimum value and a second maximum value that are each greater than the first maximum value; and transmit the generated operating state for receipt by the wireless reader.
8. The system of claim 7, further comprising: a bacteria sensor circuit disposed on the bottom surface of the flexible substrate and configured to be in electronic communication with the microcontroller.
9. The system of claim 7, wherein the moisture value is determined by obtaining a capacitance value of the at least one electrochemical moisture sensor circuit.
10. The system of claim 7, wherein the coupon is disposed on, or within a wound dressing or bandage.
11. The system of claim 7, wherein the microcontroller is configured to be powered by the radio frequency signal emitted by the transceiver of the wireless reader.
12. The system of claim 7, wherein the coupon comprises an array of electrochemical moisture sensor circuits, each in signal communication with the microcontroller.
13. The system of claim 12, further comprising: a multiplexer in electronic communication with the microcontroller, wherein the multiplexer is configured to obtain a capacitance value of each of the electrochemical moisture sensor circuits in the array of electrochemical moisture sensor circuits.
14. A system for wound care adapted for generating a wound moisture map, comprising: a flexible coupon having a top surface and a bottom surface, wherein the bottom surface is configured to be applied at least in part to a wound or ulcer; an antenna circuit and a microcontroller disposed on the

top surface of the coupon, wherein the microcontroller is in electronic communication with the antenna circuit; and a plurality of electrochemical moisture sensor circuits disposed on the bottom surface of the coupon; wherein: each electrochemical moisture sensor circuit of the plurality of electrochemical moisture sensor circuits is in electronic communication with the microcontroller; the antenna circuit is adapted to send and receive radio frequency signals to and from a remote radio frequency transceiver, respectively; the plurality of electrochemical moisture sensor circuits is adapted to be powered by radio frequency energy of the received radio frequency signals; and the controller is adapted to generate an operating state for the coupon from a set of operating states based on data from the plurality of electrochemical moisture sensor circuits, wherein the set of operating states comprises: a first operating state for the data that indicates a first moisture level between a first minimum value and a first maximum value; and a second operating state for the data that indicates a second moisture level between a second minimum value and a second maximum value that are each greater than the first maximum value.

15. The system of claim 14, further comprising an electrochemical bacteria sensor disposed on the bottom surface of the coupon that is in electronic communication with the microcontroller.

16. The system of claim 14, further comprising an electronic temperature sensor disposed on the bottom surface of the coupon that is in electronic communication with the microcontroller.

17. The system of claim 14, wherein the microcontroller is configured to: query the plurality of electrochemical moisture sensor circuits in a selective- or group-wide modality for one or more wound moisture measurements; and transmit the generated operating state to the transceiver via the antenna.

18. The system of claim 14, wherein each electrochemical moisture sensor circuit of the plurality of electrochemical moisture sensor circuits comprises an array of interdigitated electrodes.

19. The system of claim 14, further comprising a multiplexer in electronic communication with the microcontroller, wherein the multiplexer is configured to obtain a capacitance value of each of the plurality of electrochemical moisture sensor circuits.

20. The coupon of claim 1, wherein the generated operating state indicates, based on the plurality of electrochemical moisture sensors, a moisture level according to a legend scale, the legend scale comprising: a dry moisture level between the first minimum value and the first maximum value; a semi-wet moisture level between the second minimum value and the second maximum value; and a wet moisture level between a third minimum value and a third maximum value indicating more moisture than the semi-wet moisture level.
